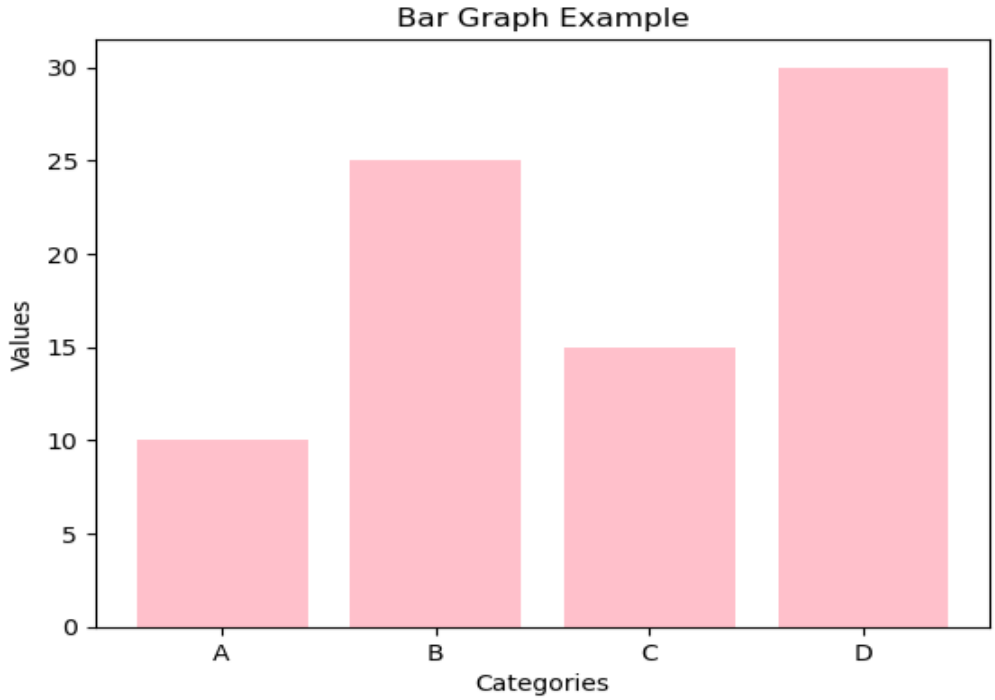
1) A chart with rectangular bars that represent data values. Bar graphs can be used to compare data or show changes over time.

```
import matplotlib.pyplot as plt
# Data
categories = ['A', 'B', 'C', 'D']
values = [10, 25, 15, 30]
# Create bar graph
plt.bar(categories, values, color='green')
# Labels and Title
plt.xlabel("Categories")
plt.ylabel("Values")
plt.title("Bar Graph Example")
# Show graph
plt.show()
```

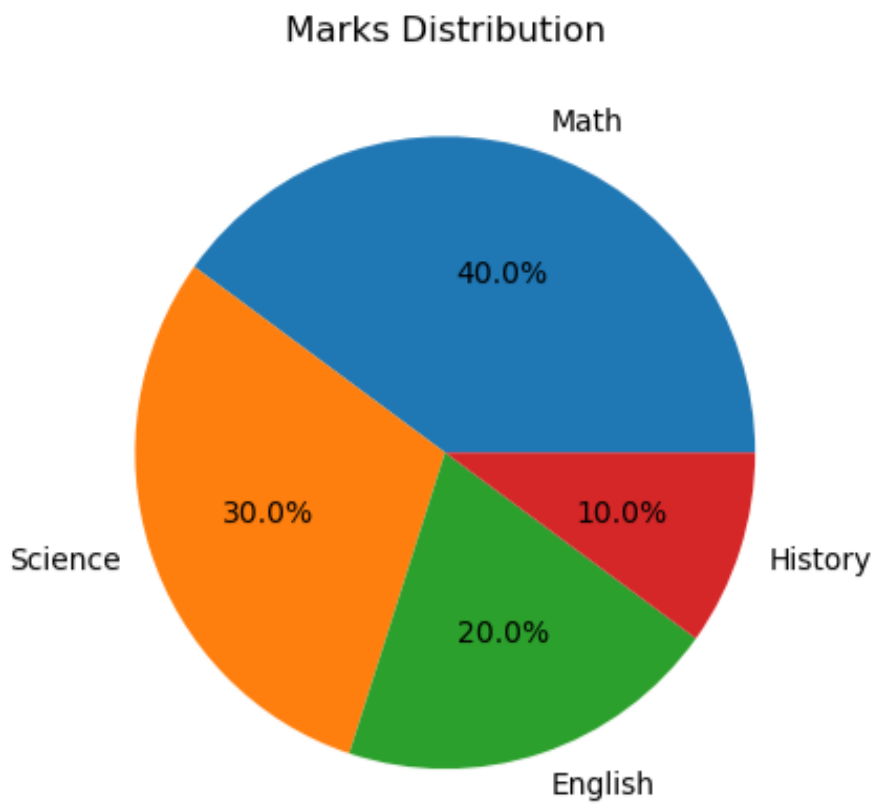
#OUTPUT:-



2) Pie chart - A circular graph with slices that represent proportions. Each slice corresponds to a category.

```
import matplotlib.pyplot as plt
# Data
subjects = ['Math', 'Science', 'English', 'History']
marks = [40, 30, 20, 10]
# Create pie chart
plt.pie(marks, labels=subjects, autopct='%1.1f%%')
# Title
plt.title("Marks Distribution")
# Show graph
plt.show()
```

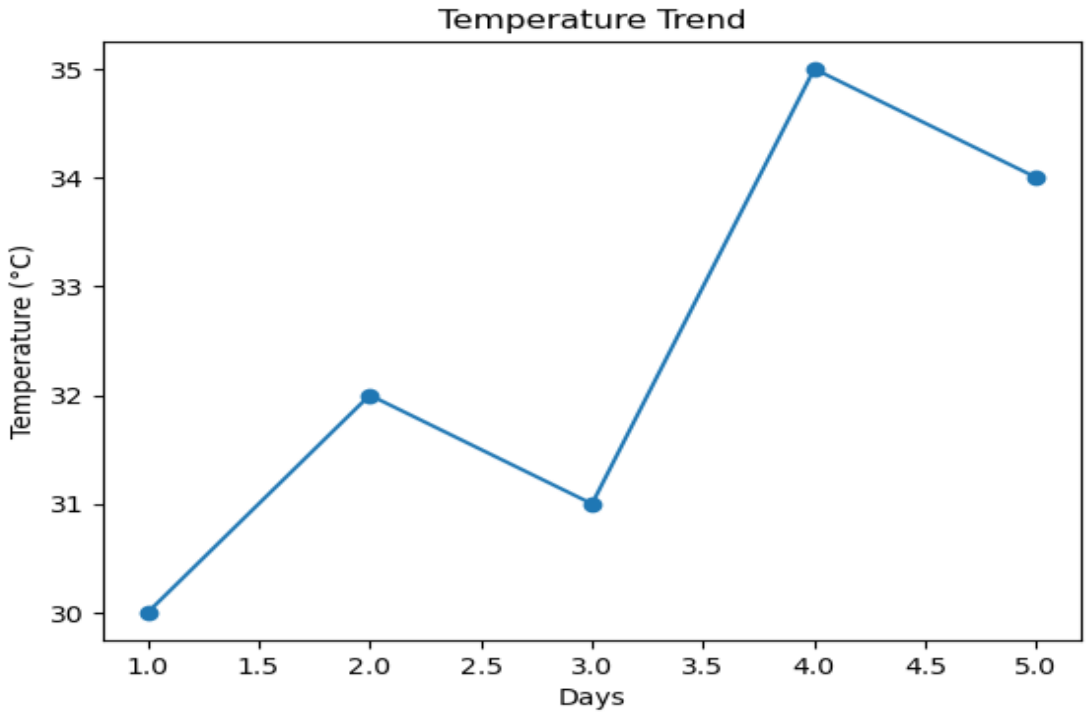
#OUTPUT:-



3) Line graph -A chart that uses lines to connect data points. Line graphs are useful for showing trends and patterns over time.

```
import matplotlib.pyplot as plt
# Data
days = [1, 2, 3, 4, 5]
temperature = [30, 32, 31, 35, 34]
# Create line graph
plt.plot(days, temperature, marker='o')
# Labels and Title
plt.xlabel("Days")
plt.ylabel("Temperature (°C)")
plt.title("Temperature Trend")
# Show graph
plt.show()
```

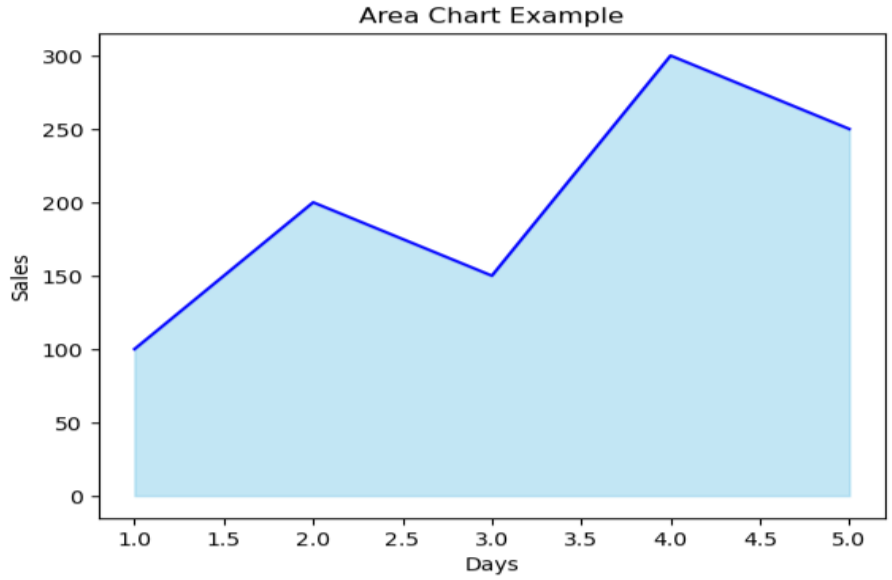
#OUTPUT:-



4) Area chart - Similar to a line chart, but with the addition of color. Area charts can be used to show cumulative totals, stacked data, and trends.

```
import matplotlib.pyplot as plt
# Data
days = [1, 2, 3, 4, 5]
sales = [100, 200, 150, 300, 250]
# Create area chart
plt.fill_between(days, sales, color="skyblue", alpha=0.5)
# Line on top (optional)
plt.plot(days, sales, color="blue")
# Labels and Title
plt.xlabel("Days")
plt.ylabel("Sales")
plt.title("Area Chart Example")
# Show graph
plt.show()
```

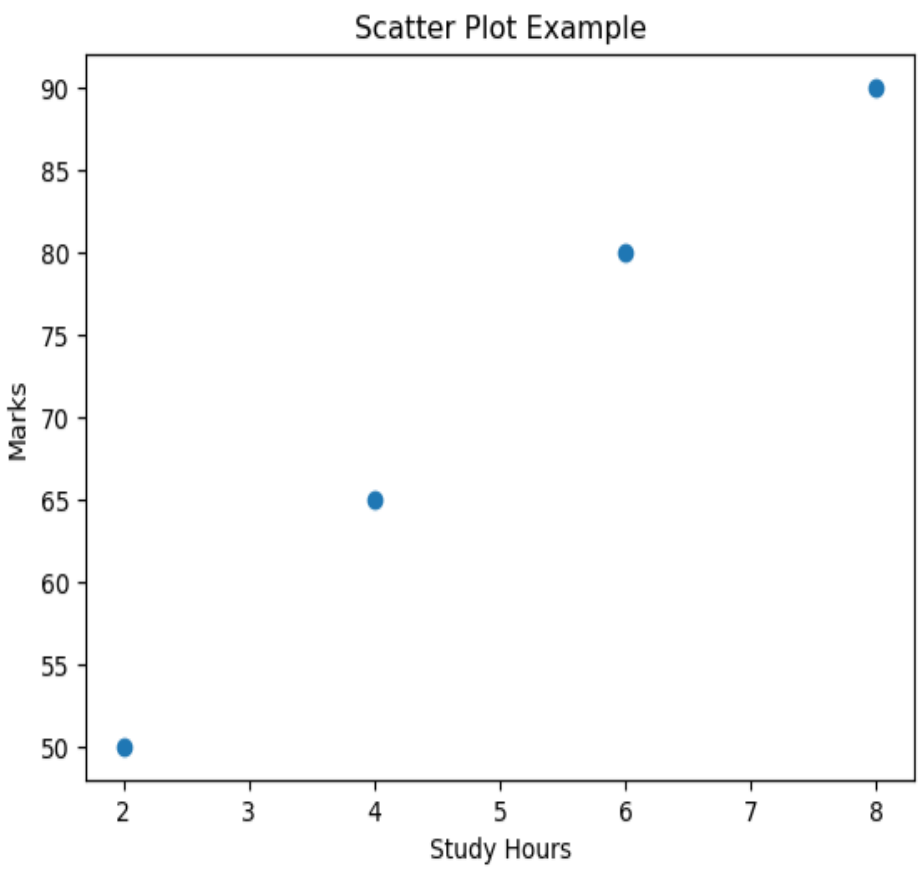
#OUTPUT:-



5) Scatter plot -A chart that uses dots to represent values for two different numeric variables.

```
import matplotlib.pyplot as plt


x = [2, 4, 6, 8] # Study hours
y = [50, 65, 80, 90] # Marks

plt.scatter(x, y)

plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Scatter Plot Example")

plt.show()
```

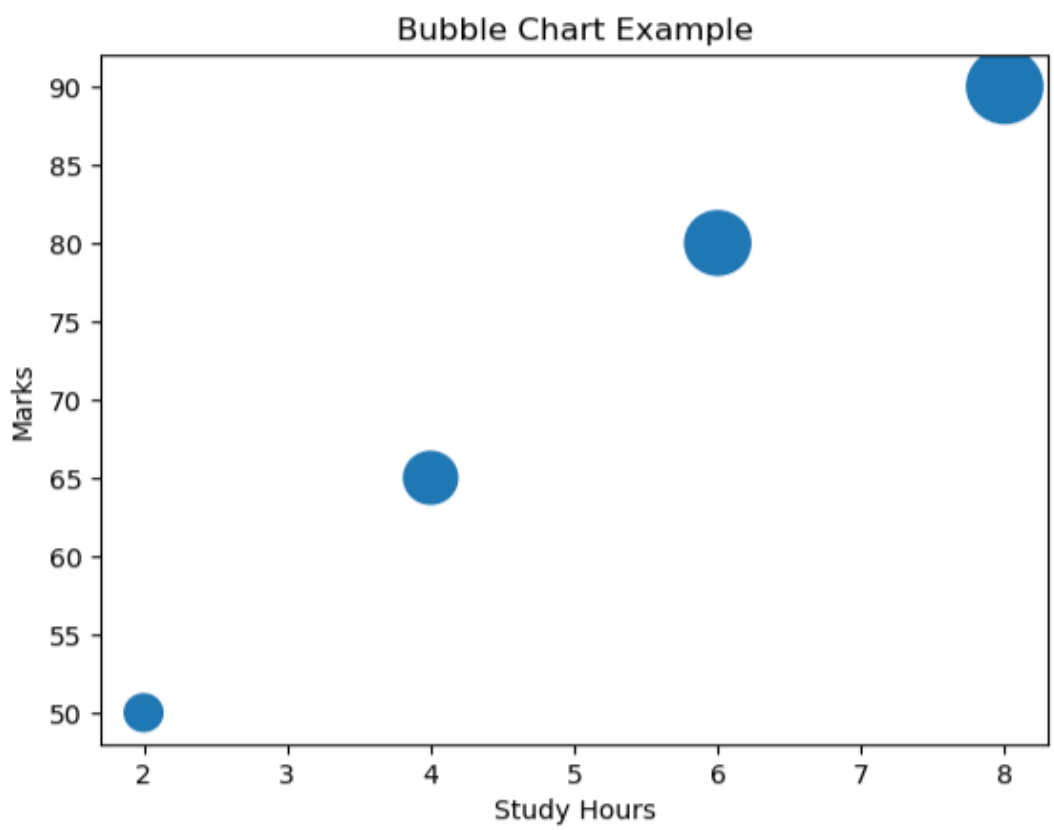
#OUTPUT:-



6) Bubble chart - A chart that displays multiple data points and their relationships.

```
import matplotlib.pyplot as plt
# Data
x = [2, 4, 6, 8] # Study Hours
y = [50, 65, 80, 90] # Marks
size = [200, 400, 600, 800] # Bubble size (Attendance)
# Create Bubble Chart
plt.scatter(x, y, s=size)
# Labels and Title
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Bubble Chart Example")
# Show chart
plt.show()
```

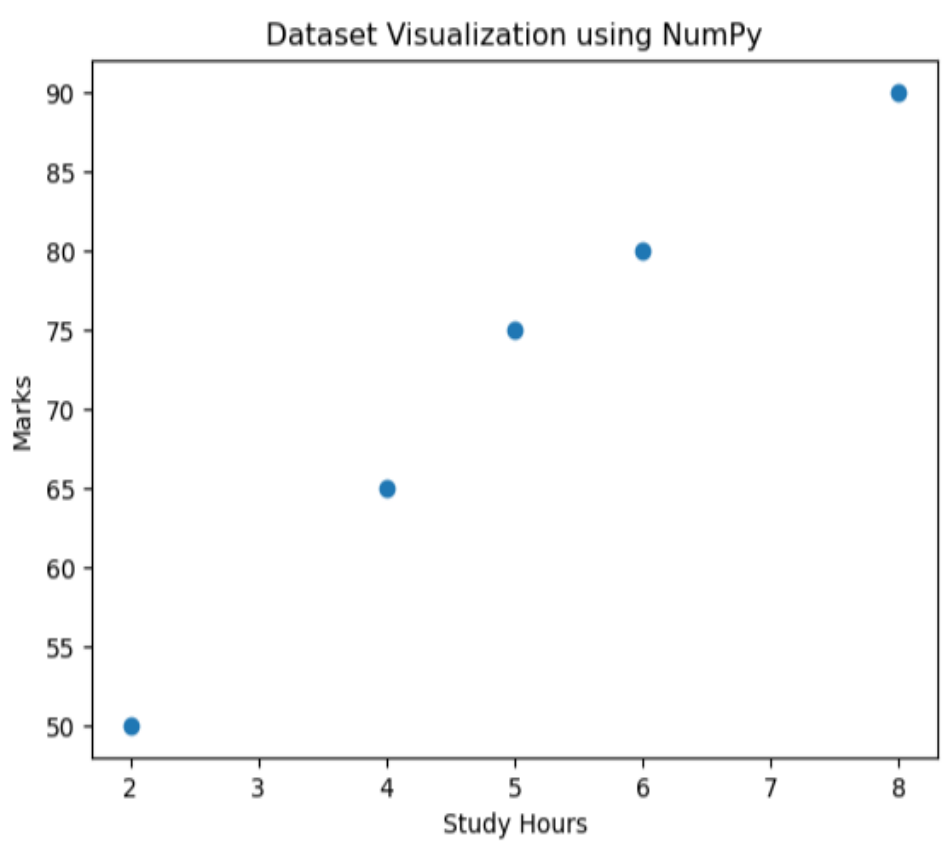
#OUTPUT:-



7) NumPy Data Visualization - Dataset for Data Visualization

```
import numpy as np
import matplotlib.pyplot as plt
# Creating dataset using NumPy
study_hours = np.array([2, 4, 6, 8, 5])
marks = np.array([50, 65, 80, 90, 75])
# Scatter plot visualization
plt.scatter(study_hours, marks)
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Dataset Visualization using NumPy")

#Display the plot
plt.show()
```

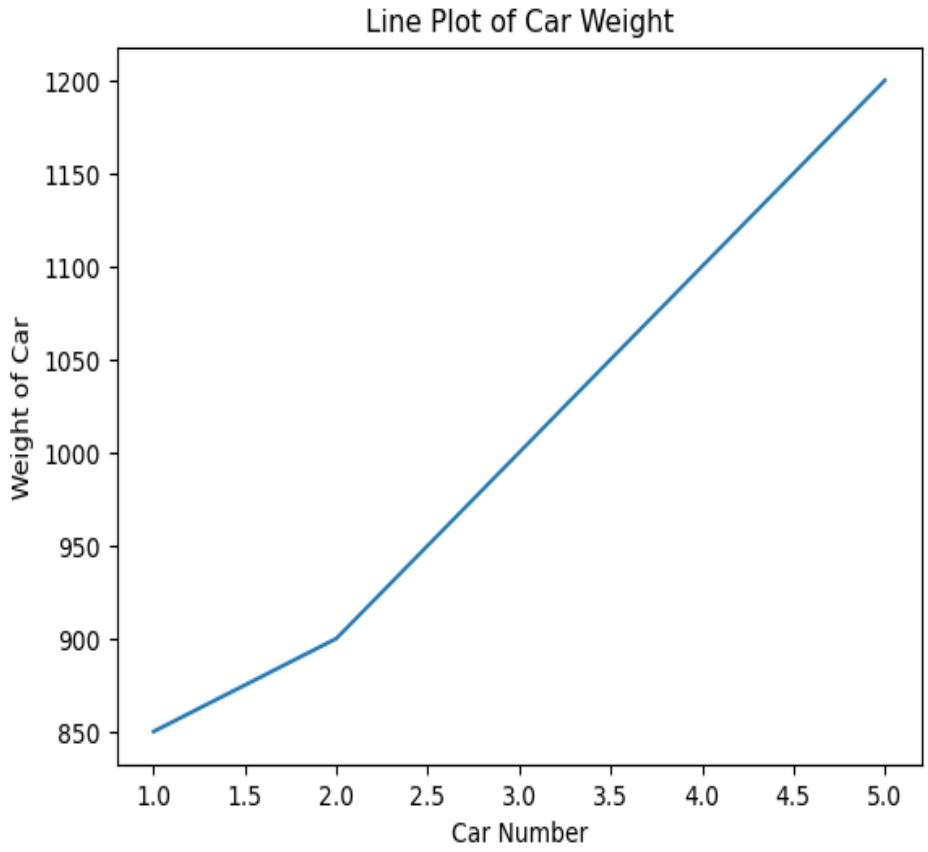
#OUTPUT:-



8) Line Plot For Data Visualization - use the plot () function to line plot the given dataset. We set the x and y coordinate of plot () as the car and weight arrays.

```
import numpy as np
import matplotlib.pyplot as plt
# Dataset
car = np.array([1, 2, 3, 4, 5])
weight = np.array([850, 900, 1000, 1100, 1200])
# Line plot
plt.plot(car, weight)
# Labels and title
plt.xlabel("Car Number")
plt.ylabel("Weight of Car")
plt.title("Line Plot of Car Weight")
# Display the plot
plt.show()
```

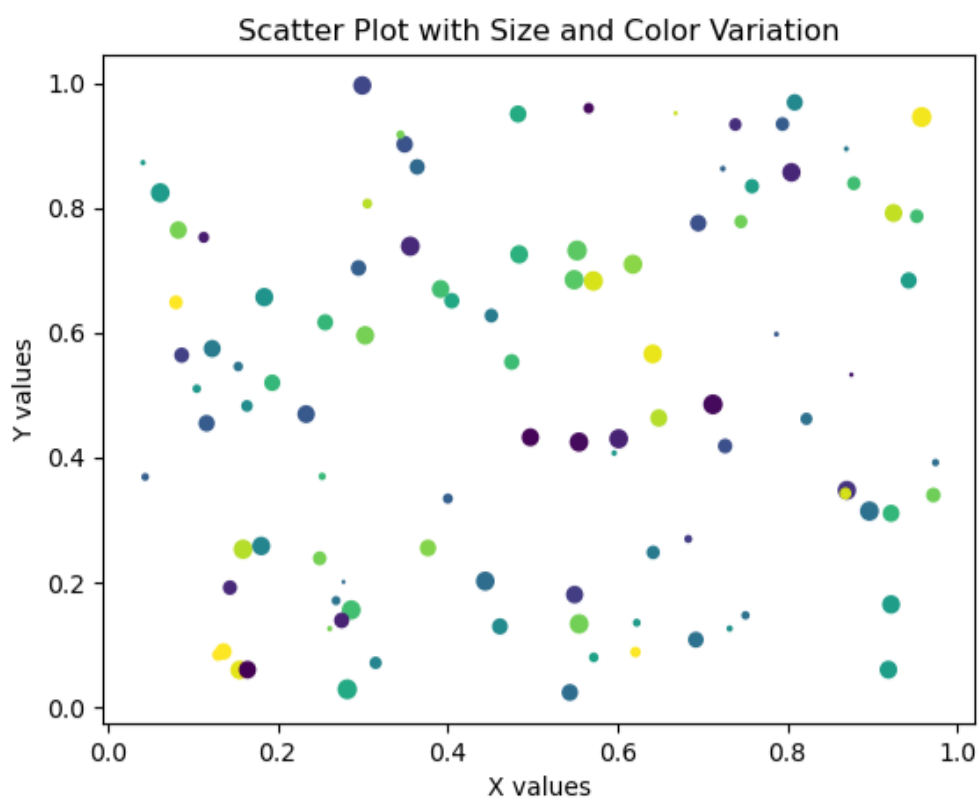
#OUTPUT:-



9) Scatter Plots for Data Visualization - Scatter Plot displays data as a collection of points. It has a scatter () function to plot the data points.

```
import matplotlib.pyplot as plt
import numpy as np
# Generate random data
x = np.random.rand(100)
y = np.random.rand(100)
sz = 50 * np.random.rand(100) # sizes of points
c = np.random.rand(100) # colors of points
# Create scatter plot
plt.scatter(x, y, s=sz, c=c)
# Labels and title
plt.xlabel("X values")
plt.ylabel("Y values")
plt.title("Scatter Plot with Size and Color Variation")
# Show plot
plt.show()
```

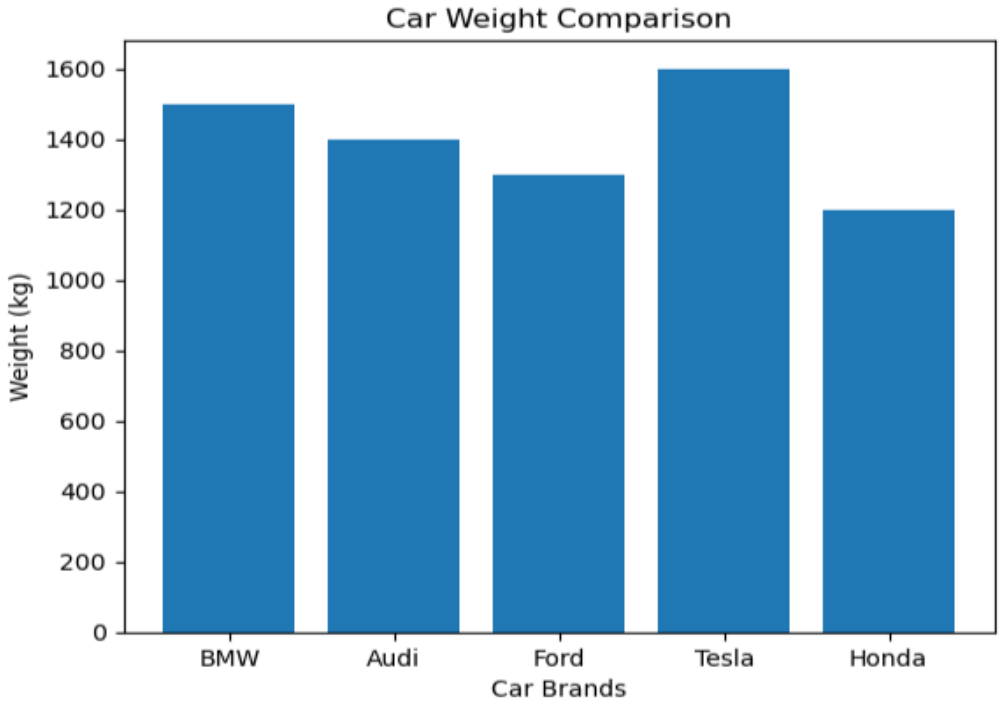
#OUTPUT:-



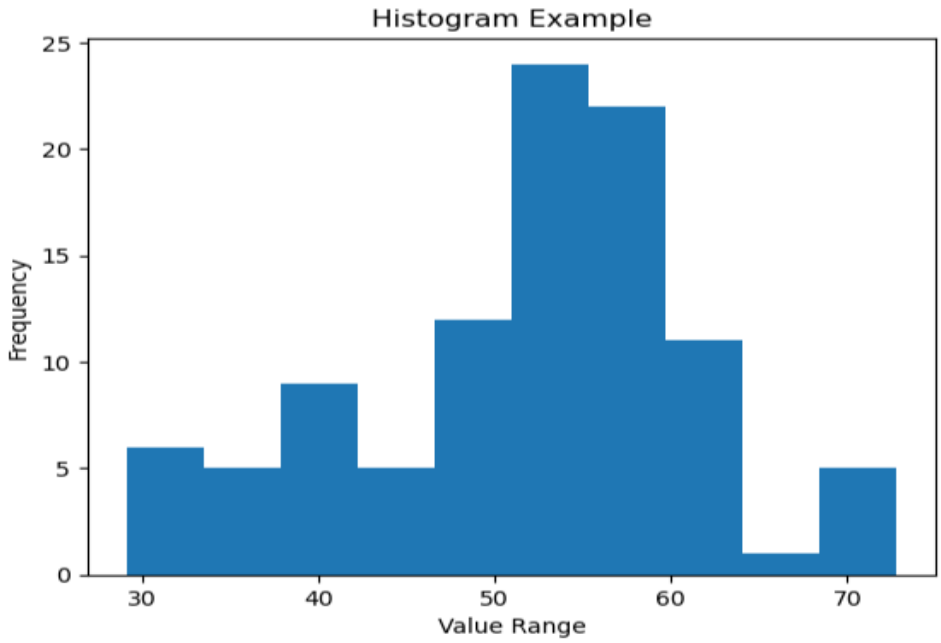
10)Bar Graphs For Data Visualization - use the bar() function to plot the bar graph and passed two arrays car and weight as its argument.

```
import matplotlib.pyplot as plt
# Data
car = ['BMW', 'Audi', 'Ford', 'Tesla', 'Honda']
weight = [1500, 1400, 1300, 1600, 1200]
# Create bar graph
plt.bar(car, weight)
# Labels and Title
plt.xlabel("Car Brands")
plt.ylabel("Weight (kg)")
plt.title("Car Weight Comparison")
# Show graph
plt.show()
```

#OUTPUT:-



11) Histograms For Data Visualization - use hist() to create histograms in NumPy. example.

```
import numpy as np

import matplotlib.pyplot as plt

# Generate random data using NumPy

data = np.random.normal(50, 10, 100)

# mean = 50, standard deviation = 10, 100 values

# Create histogram

plt.hist(data, bins=10)

# Labels and Title

plt.xlabel("Value Range")

plt.ylabel("Frequency")

plt.title("Histogram Example")

# Show graph

plt.show()
```

#OUTPUT:-